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Subject Name :Data Structures

Subject code:0316

ASSESSMENT TOLL-C2(AT-1)

37. Assume, a single character is present in the data field of DLL with 6 nodes as

S<=>I<=>M<=>A<=>T<=>S. Write a function to print the same in reverse

order.

```
#include <stdio.h>
#include <stdlib.h>

struct Node
{
   char data;
   struct Node *prev;
   struct Node *next;
};

// Function to create a new node
struct Node* createNode(char ch)
{
   struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
   newNode->data = ch;
   newNode->prev = NULL;
   newNode->next = NULL;
```

```
    return newNode;
}

// Function to print DLL in reverse order
void printReverse(struct Node *head)
{
    struct Node *temp = head;

    // Move to the last node
    while(temp->next != NULL)
        temp = temp->next;

    printf("Reverse Order: ");

    // Traverse backwards
    while(temp != NULL)
    {
        printf("%c ", temp->data);
        temp = temp->prev;
    }
}

int main()
{
    struct Node *head, *n2, *n3, *n4, *n5, *n6;

    head = createNode('S');
    n2 = createNode('I');
    n3 = createNode('M');
    n4 = createNode('A');
    n5 = createNode('T');
    n6 = createNode('S');
```

```
    head->next = n2;
    n2->prev = head;
    n2->next = n3;
    n3->prev = n2;
    n3->next = n4;
    n4->prev = n3;
    n4->next = n5;
    n5->prev = n4;
    n5->next = n6;
    n6->prev = n5;

    printReverse(head);

    return 0;
}
```

Assume a single character is present in the data field of a Single Linked List with 6 nodes as S- > I -> M -> A -> T -> S. Write a function to print the same in reverse order using stack data structure.

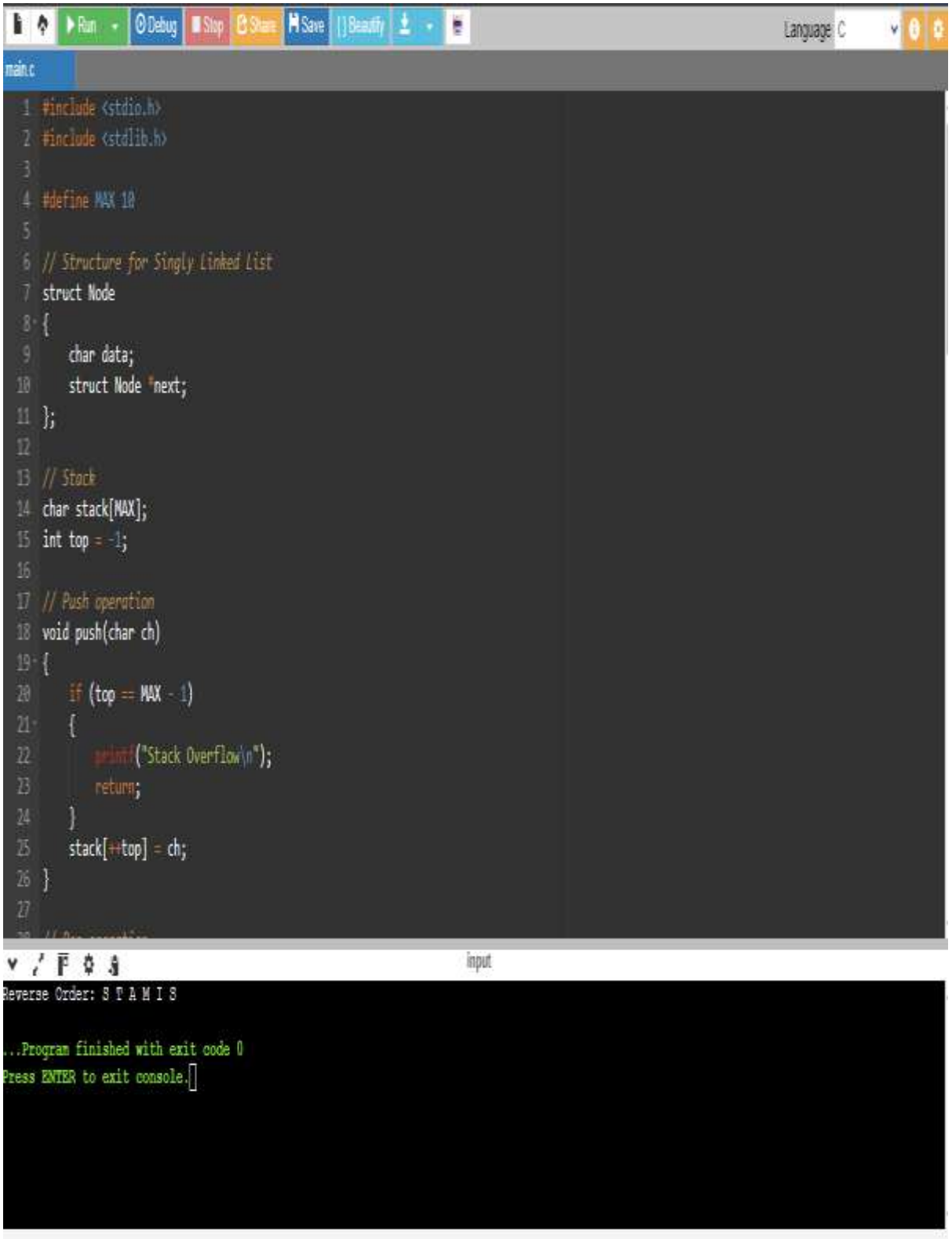

Run
Debug
Stop
Share
Save
{} Beautify
Language C
main.c
#include <stdio.h>
#include <stdlib.h>
struct Node
{
char data;
struct Node *prev;
struct Node *next;
};
// Function to create a new node
struct Node* createNode(char ch)
{
struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
newNode->data = ch;
newNode->prev = NULL;
newNode->next = NULL;
return newNode;
}
// Function to print DLL in reverse order
void printReverse(struct Node *head)
{
struct Node *temp = head;
// Move to the last node
while(temp->next != NULL)
input
Reverse Order: S T A M I S
...Program finished with exit code 0
Press ENTER to exit console.

38.Assume a single character is present in the data field of a Single Linked List with 6 nodes as S- > I -> M -> A -> T -> S. Write a function to print the same in reverse order using stack data structure.

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 10

// Structure for Singly Linked List
struct Node
{
   char data;
   struct Node *next;
};

// Stack
char stack[MAX];
int top = -1;

// Push operation
void push(char ch)
{
   if (top == MAX - 1)
   {
      printf("Stack Overflow\n");
      return;
   }
   stack[++top] = ch;
}
```

```
// Pop operation
char pop()
{
    if (top == -1)
    {
        return '\0';
    }
    return stack[top--];
}

// Function to print linked list in reverse using stack
void printReverse(struct Node *head)
{
    struct Node *temp = head;

    // Push all node data into stack
    while (temp != NULL)
    {
        push(temp->data);
        temp = temp->next;
    }

    // Pop and print
    printf("Reverse Order: ");
    while (top != -1)
    {
        printf("%c ", pop());
    }
}

// Main Function
int main()
```

```
{
    struct Node *head, *n2, *n3, *n4, *n5, *n6;

    // Create nodes
    head = (struct Node *)malloc(sizeof(struct Node));
    n2 = (struct Node *)malloc(sizeof(struct Node));
    n3 = (struct Node *)malloc(sizeof(struct Node));
    n4 = (struct Node *)malloc(sizeof(struct Node));
    n5 = (struct Node *)malloc(sizeof(struct Node));
    n6 = (struct Node *)malloc(sizeof(struct Node));

    // Store data
    head->data = 'S';
    n2->data = 'I';
    n3->data = 'M';
    n4->data = 'A';
    n5->data = 'T';
    n6->data = 'S';

    // Link nodes
    head->next = n2;
    n2->next = n3;
    n3->next = n4;
    n4->next = n5;
    n5->next = n6;
    n6->next = NULL;

    // Print in reverse
    printReverse(head);

    return 0;
}
```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Language C
main.c
#include <stdio.h>
#include <stdlib.h>
#define MAX 10
// Structure for Singly Linked List
struct Node
{
char data;
struct Node *next;
};
// Stack
char stack[MAX];
int top = -1;
// Push operation
void push(char ch)
{
if (top == MAX - 1)
{
printf("Stack Overflow\n");
return;
}
stack[++top] = ch;
}
input
Reverse Order: S T A M I S
...Program finished with exit code 0
Press ENTER to exit console.